

SHAWN AGARWAL

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EDUCATION

University of Oklahoma, B.S. Electrical Engineering

Expected May 2027

GPA: 4.0 • Oklahoma Regents Scholar • Presidential Honor Roll • Distinguished Freshman Award

Norman, OK

Relevant Coursework: Signals & Systems, Electromagnetic Fields & Wave Propagation, Solid-State Electronic Devices, Digital Design Laboratory, PCB Design Principles, Energy Conversion

EMBEDDED & HARDWARE ENGINEERING

Intelligent Ground Vehicle Competition, Electrical Member

Aug 2025 - May 2026

- Designed, built, and brought up custom PCBs with power regulation, signal conditioning, and hardware safety interlocks for fault-tolerant sensor integration, owning schematic capture through layout, assembly, and system-level verification against the competition safety requirements
- Wrote real-time embedded control firmware in C/C++ on STM32 (ARM Cortex-M) under an RTOS, using prioritized task scheduling to keep closed-loop actuation deterministic; integrated LiDAR, cameras, GPS, IMU, encoders, and e-stop hardware on a single CAN bus and built a CAN-to-USB bridge for live bus monitoring and fault reproduction
- Wrote and ran bench and vehicle-level test procedures, isolating hardware and firmware faults with oscilloscope and logic-level probing and closing them out inside fixed competition build deadlines

Student Tele-Operated Robotics Mission, Electrical Team Lead

Aug 2024 - May 2025

- Designed multi-voltage power rails and buck converters to a **<5% ripple** specification for reliable motor and sensor supply; implemented IR/RF signal decoding integrated with embedded firmware for remote command and control
- Led the electrical team through design reviews, subsystem integration, and structured test phases, tracking requirements verification against a fixed end-of-year mission schedule

VEX Robotics Competition, Team Lead — Canadian Valley Technology Center

Aug 2023 - May 2024

- Led end-to-end robot design, build, and test across iterative competition cycles; qualified for and competed at the VEX Robotics World Championship

EXPERIENCE

Operations Intern, Amazon, OKC1 Fulfillment Center

May 2026 - Aug 2026

Robotics-driven fulfillment operations

Oklahoma City, OK

- Analyzed process and flow data to find throughput bottlenecks, and modeled a process-flow change projected to save **~\$800K annually** at the site
- Ran a **280-associate** operation on the floor alongside technicians, engineers, and area managers, enforcing safety, quality, and operational-readiness standards through daily production targets

Undergraduate Researcher, Engineering Pathways Fellow

Jan 2026 - Present

Center for Quantum Research & Technology (CQRT), University of Oklahoma

Norman, OK

- Grow and characterize epitaxial oxide and nitride thin films and heterostructures, running vacuum deposition and measurement hardware in Dr. Hanjong Paik's group
- Evaluate BaTiO₃ / SrTiO₃ / Si heterostructures by comparing high-resolution X-ray diffraction against transport and optical measurements to assess crystalline quality, strain state, and interface behavior
- Wrote XRDC, a MATLAB tool for HRXRD analysis with rocking-curve and reciprocal-space workflows, replacing manual peak fitting and standardizing film-quality metrics across the group

Founder & Engineer, Heading Software

Jan 2026 - Present

AI document intelligence for oil & gas land services

Oklahoma

- Built an LLM document-extraction and verification pipeline (OCR, schema validation, citation grounding) end to end, with an offline evaluation harness tracking per-field precision and recall against a hand-labeled test set
- Wrote the BullMQ/Redis batch layer with idempotent workers and automated QA catching drift, duplicates, and corrupt inputs

Undergraduate Researcher, Machine Learning

Sep 2024 - May 2025

Data Institute for Societal Challenges (DISC), University of Oklahoma

Norman, OK

- Fine-tuned BERT for relevance classification using active-learning uncertainty sampling over **1.6M records** for retrieval ranking

SKILLS

Hardware: Schematic capture & PCB layout (Altium), power electronics, buck converters, signal conditioning, STM32 / ARM Cortex-M, RTOS, CAN/FDCAN, SPI, I²C, UART, oscilloscope/DMM bring-up and debug, soldering and board rework, Fusion 360

Software: C/C++, Python, MATLAB, Rust, SQL, PyTorch, TensorFlow, Git, LaTeX

Test & Analysis: High-resolution X-ray diffraction, transport measurement, optical characterization, vacuum deposition systems, MATLAB data analysis

Activities: IEEE, Engineers' Club, Sooner Competitive Robotics, Society of Asian Scientists and Engineers